Ensuring fair access to online education platform for remote communities

D4F Assignment

Use case: Ensuring fair access to online education for remote communities

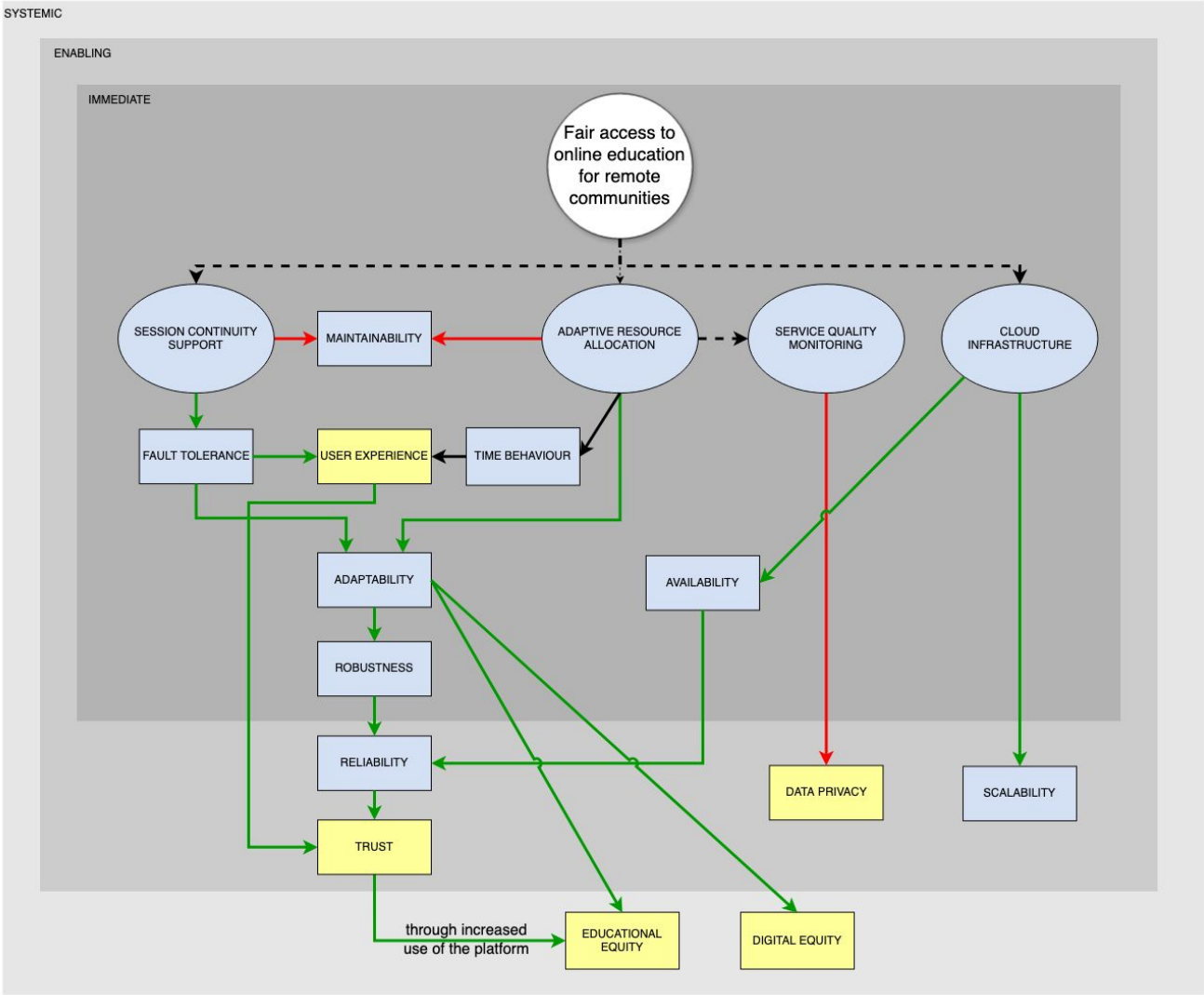
- **Persona**: Rohan, 17 years old, lives in a rural village in India, attends high-school classes online on a global educational platform, as he is unable to attend school physically due to the remote location of his village. He is using a shared laptop and relies on mobile internet access (3G) for accessing the internet. His internet connection is unstable, he is experiencing fluctuating network performance and frequent network outages.

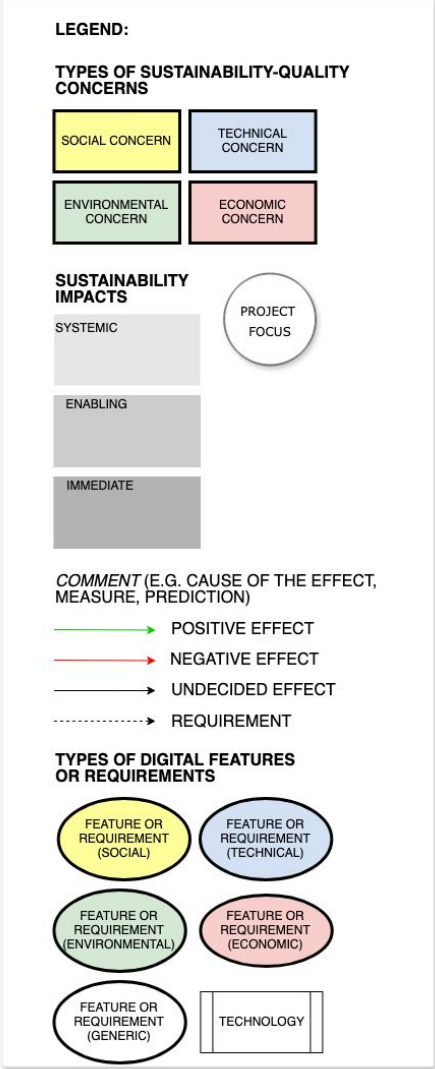
- **Context**: The educational platform is hosted on a global cloud infrastructure with an edge network, offering online classes for high-school students. The platform offers digital resources, on-demand and real-time online classes, and has an assessment component as well, offering both self-paced and real-time proctored virtual tests. Connectivity issues and poor network conditions frequently lead to session interruptions, significantly impacting Rohan's learning experience and progress.

- **Scenario:** Rohan is ready to take an important real-time proctored test necessary for his advancement to the next academic year. During his real-time proctored exam, Rohan logs in and relies on a stable Internet connection for identity verification, monitoring, and submission. Soon after he starts, network fluctuations cause delays, video lag, and disconnections, disrupting his exam-not because of his ability, but because of poor connectivity.

- **Proposed solution**: The architecture will address network issues and aim at counter-balancing the disadvantage caused by socioeconomic circumstances, with features designed to ensure fair access.

Definition of Fairness: Decision Map





Definition of Fairness: Decision Map (cont.)

To contribute to fair access to the online education platform by offering comparable quality of service to all users, including users based in remote locations, who often face chronic network instability and poor network quality, the following technical features are relied upon:

- ***cloud infrastructure***:
    - deploy cloud and edge nodes strategically, with edge nodes located close to areas with network instability
    - caching frequently accessed and critical content in edge nodes and serving users from the edge nodes, thereby providing shorter response times and reduced latency
    - contributes to *scalability*, *availability*, and indirectly to *reliability* of the platform
- ***adaptive resource allocation***:
    - dynamically assign cloud and edge computing- and network resources in real-time, depending on current system conditions, service quality factors of users such as network instability or connectivity issues and user context, such as the priority of the ongoing task (critical tasks e.g. exams are handled with priority)
    - has a significant impact on the fairness perspective through supporting *educational*- and *digital equity*
    - relies on ***service quality monitoring***:
        - continuous monitoring of service quality
        - has a negative impact on *data privacy*, due to tracking sensitive information implying socioeconomic circumstances of users
        - prioritization logic based on user context and resource availability
        - triggering *adaptive resource allocation* workflows to accommodate users experiencing degraded service quality
- ***session continuity support****:*
    - prediction of users' needs (e.g., scheduled classes or exams) and proactively caching necessary resources in close-by edge nodes to avoid session interruptions

The proposed features indirectly build user trust, thereby increasing platform usage and improving educational equity, creating a positive feedback loop that supports fair access to education. These features minimize geographic and infrastructure barriers, enabling fuller participation from remote communities. Additionally, improved adaptability has a positive impact on digital equity, allowing access to online resources to a larger group of people, from diverse socioeconomic backgrounds.

Note on trade-off: The adaptive resource allocation creates a notable trade-off. Time behavior and user experience are expected to improve for users with disadvantaged network access, but other users may experience degraded service due to resources being dedicated elsewhere, hence, the effect is marked as "undecided" (black arrow).

Definition of Fairness: SQ model

QUALITY ATTRIBUTE (QA)	DEFINITION	REFERENCE	TECHNICAL	ENVIRONMENTAL	ECONOMIC	SOCIAL	DEFINITION of METRIC/KPI
MAINTAINABILITY	degree of effectiveness and efficiency with which a product or system can be modified by the intended maintainers	[ISO/IEC 25010:2011]	X				Average time to resolve system faults.
FAULT TOLERANCE	degree to which a system or component operates as intended despite the presence of hardware or software faults	[ISO/IEC 25010:2011]	X				Session recovery success rate, calculated as the number of interrupted sessions resumed successfully divided by the total number of interrupted sessions.
TIME BEHAVIOUR	degree to which the response, processing times and throughput rates of a system, when performing its functions, meet requirements	[ISO/IEC 25010:2011]	X				Average network latency, calculated as total accumulated latency divided by number of requests.
ADAPTABILITY	degree to which a system can effectively and efficiently be adapted for different or evolving hardware, software or other opetational or usage environments	[ISO/IEC 25010:2011]	X				Average delay from detection of changing resource demands to adaptive resource reallocation completion.
ROBUSTNESS	the degree to which a sytem behaves in an acceptable way in unexpected situations	[S-Cube Quality Reference Model (2012)]	X				Rate of successful auto-recoveries during critical tasks.
RELIABILITY	degree to which a system, product or component fulfils certain functions under certain conditions over a certain period of time.	[ISO/IEC 25010:2011]	X				Average time between failures that directly impact end-user experience, including outages or service degradation.
TRUST	degree to which a user or other stakeholder has confidence that a product or system will behave as intended	[ISO/IEC 25010:2011]				X	Percentage of users who would recommend the platform, based on monthly user survey results.
USER EXPERIENCE	all aspects of the user's perceptions and responses resulting from the use and/or anticipated use of a product, system, or service	[ISO 9241-210:2019]				X	Average perceived usability score based on monthly user survey results on a 10-point scale.
AVAILABILITY	degree to which a system or component is operational and accessible when required for use	[ISO/IEC 25010:2011]	X				Annual platform uptime percentage, typically calculated over a rolling 12-month period.
SCALABILITY	capability of a product to handle growing or shrinking workloads or to adapt its capacityto handle variability	[ISO/IEC 25010:2011]	X				Number of concurrent sessions the platform can support above defined service quality levels.
DATA PRIVACY	degree to which individuals, groups, or institutions determine for themselves when, how, and to what extent information about them is communicated to others	A F Westin. "Privacy and freedom". New York: Atheneum Press (1967)				X	no applicable KPI
EDUCATIONAL EQUITY	degree to which personal or social circumstances such as gender, ethnic origin or family background, are not obstacles to achieving educational potential and that all individuals reach at least a basic minimum level of skills	extended from: OECD. (2012). Equity and Quality in Education: Supporting Disadvantaged Students and Schools. OECD Publishing				X	Annual increase in the percentage of students successfully finishing their academic year, specifically focusing on users from disadvantaged backgrounds.
DIGITAL EQUITY	degree to which digital society empowers all citizens through equal opportunities to access and use ICTs for economic and social development regardless of location, socio-economic status, age, gender, culture, language, or individual capabilities	extended from: International Telecommunication Union (ITU). (2020). Measuring Digital Development: Facts and Figures 2020. ITU Publications				X	Annual increase in the percentage of enrolled students from underrepresented or disadvantaged groups.

Definition of Fairness: SQ model (cont.)

The quality attributes are mapped to specific sustainability dimensions based on their fundamental characteristics and primary impacts.

Technical quality attributes (*maintainability, fault tolerance, time behaviour, adaptability, robustness, reliability, availability, scalability*) are classified as technical because they directly relate to the system's functional capabilities, engineering design, and operational performance. For example, fault tolerance is technical because it concerns the system's inherent ability to continue operating despite hardware or software failures, while scalability addresses the core technical capacity to handle varying workloads.

Social quality attributes (*trust, user experience, data privacy, educational equity, digital equity*) are classified under the social dimension because they primarily concern human experiences, rights, and societal impacts. Trust and user experience focus on human perception and interaction, data privacy addresses fundamental rights to information control, while educational and digital equity directly target societal inclusion and accessibility for diverse populations.

Due to the use case and the selected fairness definition, the decision map and SQ model target systematic improvements to social quality attributes (*educational equity*, *digital equity*), which - as per the proposed solution - are to be achieved through targeting technical quality attributes first. As the decision map highlights, by addressing technical quality attributes in the scope of the project, with immediate sustainability impact - indirectly impacting user experience positively, which is regarded as a social concern due to the focus on the perception of users - technical, environmental and social quality attributes are impacted mid-term (enabling impact) and long-term (systemic impact).

Environmental and economic quality attributes could be attributed to the proposed solution as well, but are out of scope for the purposes of the assignment and the selected fairness perspective. Hence, these are not integrated into the decision map and SQ model.

Fairness perspective

DESIRED QUALITY	The equitable provision of online educational resources, learning opportunities, and assessment conditions across diverse geographic and socioeconomic contexts, ensuring that technical constraints and infrastructure limitations do not systematically disadvantage students in remote areas. Through adaptive network traffic priority scheduling, the system improves educational equity, digital equity, and overall user trust, ensuring a resilient and inclusive learning environment.
CONCERNS	-**Adaptability**: The system's ability to dynamically adjust to network conditions and resource constraints, ensuring continuous access to online classes and proctored exams even in low-connectivity environments. -**Fault tolerance**: The system's ability to ensure uninterrupted access to online learning and proctored exams by implementing error recovery, redundancy, and failover mechanisms, preventing disruptions due to network instability, system failures, or connectivity issues. -**Time behavior**: The system's responsiveness in delivering real-time learning and assessments, minimizing latency and processing delays to ensure seamless exam monitoring and uninterrupted interactions.
ARCHITECTURAL TACTIC(S)	1. **Prioritized Traffic Management**: Implement Quality of Service-based scheduling to ensure that important learning activities, such as real-time proctored exams, receive priority network resources over less time-sensitive tasks. [1] 2. **Dynamic Resource Allocation**: Allocating computational, storage, and network resources dynamically based on real-time demand, user needs, and system constraints. [2] 3. **Edge Caching**: Deploying proactive and predictive caching strategies at edge nodes near remote users, by storing frequently accessed content locally. [2]
POSSIBLE KPI(S)	**Access to reliable digital learning for students=** f(session continuity, tasks completion, network latency)
PROBLEMS AND PITFALLS	**1. Privacy and Trust:** Monitoring session quality and user connectivity requires data collection that may raise privacy concerns [3] **2. Limited Edge Node Availability:** Some rural users may not have access to nearby edge nodes, reducing the system's effectiveness [4] **3. Fairness Trade-offs:** Allocating resources to users with poor connectivity may lead to reduced service quality for users with stable internet [3]

[1] White, R., & Banks, E. (2018). Computer networking problems and solutions. Pearson Education. pp. 195-220.
[2] Bass, L., Clements, P., & Kazman, R. (2021). Software architecture in practice (4th ed.). Addison-Wesley Professional. pp. 219-220.
[3] Arroba, P., Buyya, R., Cárdenas, R., Risco-Martín, J. L., & Moya, J. M. (2023). Sustainable edge computing: Challenges and future directions. Journal of Software: Practice and Experience.
[4] Saini, K., Pandey, U., & Raj, P. (2022). Edge computing challenges and concerns. In P. Raj, K. Saini, & C. Surianarayanan (Eds.), Advances in Computers (Vol. 127, pp. 259–278). Elsevier

Definition of KPI*

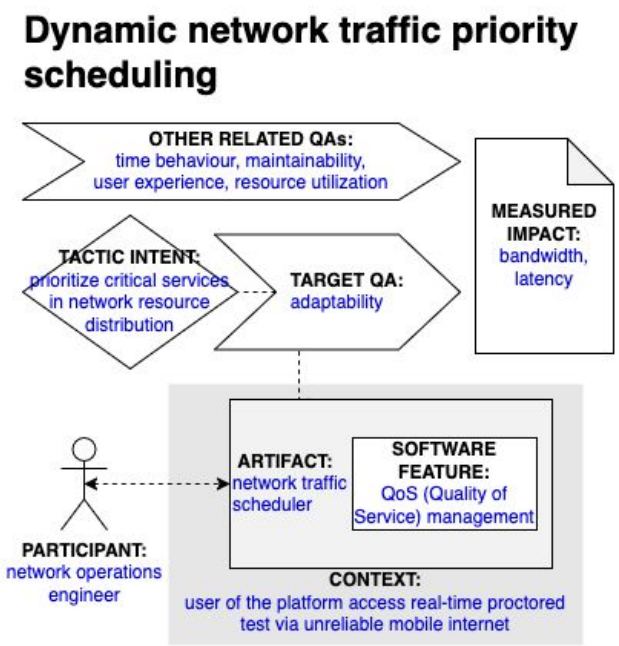
KPI Name: Reliable digital learning for students					
Goal	**Critical Success Factor (CSF)**	**Key Performance Indicator (KPI)**		**Metric(s)**	**Measure(s)**
Make educational platforms accessible, inclusive, and reliable for all users regardless of connectivity constraints [1].	***By next year, all students should have reliable access to digital learning that meets agreed targets.***	***Access to reliable digital learning for students=*** ***f(session continuity, tasks completion, network latency)***		***Session continuity=*** ***f(uninterrupted sessions / total sessions) [4]*** ***Tasks completion =*** ***f(successfully completed learning activities / attempted learning activities) [5]*** ***Network latency = f(total latency time / number of requests) [4]***	***- Number of learning sessions attempted (#)*** ***-Number of uninterrupted sessions (#)*** ***-Number of attempted learning activities (#)*** ***-Number of successfully completed learning activities (#)*** ***- Network latency per session interaction (ms)*** ***-Number of requests (#)***
		Target	**Action(s)**		
		Achieve a: ***-Session continuity ≥ 95%*** ***-Task completion ≥ 85% [2]*** ***-Average latency ≤ 150 ms [3]*** ***and an overall KPI Score ≥ 0.85, based on equally weighted values, where network latency is converted into a normalized score (0-1)***	***Improve relevant factors, e.g., reduce latency through edge nodes placed near users and optimizing content with adaptive delivery for low-bandwidth access.***		

[1] Braun, A., März, A., Mertens, F., & Nisser, A. (2020). Rethinking education in the digital age. https://doi.org/10.2861/84330
[2] Koenig, R. (2019, June 6). Moving from 5% to 85% completion rates for online courses. EdSurge. https://www.edsurge.com/news/2019-06-06-moving-from-5-to-85-completion-rates-for-online-courses (Accessed March 23, 2025)
[3] Suznjevic, M., & Matijasevic, M. (2014). Network traffic characterization and classification of online games. In L. T. Yang, A. Y. Zomaya, & S. Y. Kung (Eds.), Encyclopedia of Cloud Computing (pp. 1–10). Springer. https://doi.org/10.1007/978-981-4560-52-8_23-1
[4] International Organization for Standardization. (2011). ISO/IEC 25010:2011: Systems and software engineering — Systems and software Quality Requirements and Evaluation (SQuaRE) — System and software quality models.
[5] International Organization for Standardization. (2016). ISO/IEC 25023:2016: Systems and software engineering — Systems and software Quality Requirements and Evaluation (SQuaRE) — Measurement of system and software product quality.

Architectural Tactic(s) for Fairness*

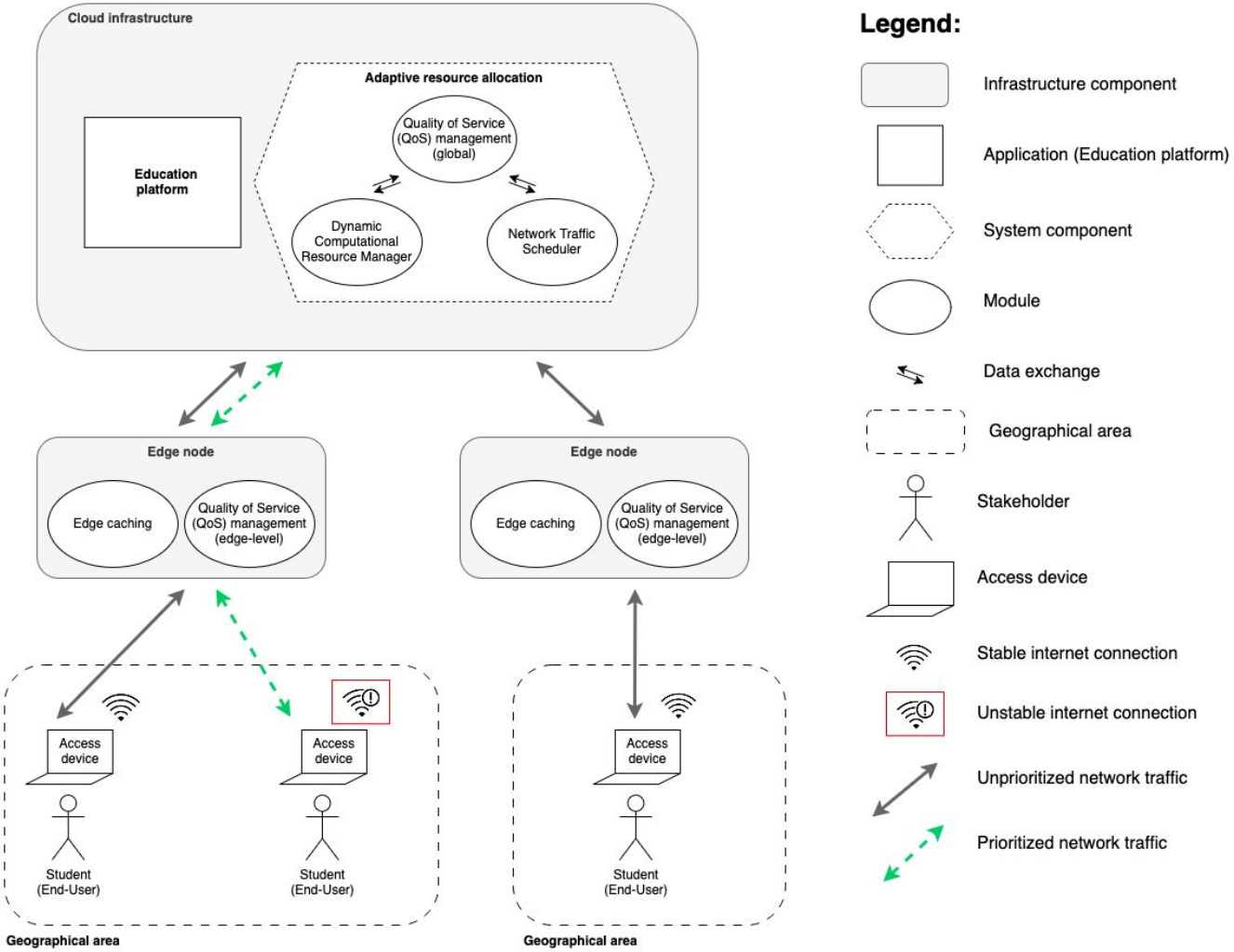
Proposed tactics to address fairness perspective:

- Dynamic resource allocation
- **Dynamic network traffic priority scheduling => *selected for elaboration, due to most direct relation to use case***
- Edge caching



TITLE	Dynamic network traffic priority scheduling
DESCRIPTION	*Dynamically adjusts the priority of network traffic flows in real-time by the network traffic scheduler, based on data from the QoS management system, with the aim to proactively identify and prioritize users executing critical tasks, who are experiencing degraded service quality due to unstable or poor-quality internet connection and are therefore in need of optimized bandwidth allocation.*
PARTICIPANT	*network operations engineer*
ARTIFACT	network traffic scheduler
CONTEXT	users executing critical tasks with access with poor network quality and/or network interruptions (e.g. access real-time proctored test via unreliable mobile internet)
SOFTWARE FEATURE	Quality of Service management (QoS)
TACTIC INTENT	prioritize critical services in network resource distribution
TARGET QUALITY ATTRIBUTE	adaptability
OTHER RELATED QUALITY ATTRIBUTES	time behaviour, maintainability, user experience, resource utilization
MEASURED IMPACT	bandwidth, latency
SOURCES	- extended to the network context from: Len Bass, Paul Clements, and Rick Kazman (2021) Software Architecture in Practice, 4th Edition. Addison-Wesley Professional, p.219-220 - see also: Russ White and Ethan Banks (2018). "Computer Networking Problems and Solutions. Pearson Education. p.195-220.

Fairness view: Fair access to online education platform for remote communities



Fairness view (cont.)

The Education platform is hosted in the cloud. The infrastructure also comprises a distributed network of edge nodes.

Students accessing the platform - rather than repeatedly fetching from centralized cloud resources - interact with the education platform through the edge node closest to them, shortening the path for the network traffic between the platform and its users by serving users from the edge nodes geographically close to them, which is especially relevant for users accessing the platform from remote communities with unstable internet access.

An additional aspect of ensuring fair access for such users, is managed by the cloud infrastructure system component *adaptive resource allocation*, consisting of multiple modules. The *QoS (Quality of Service) management* module continuously monitors service quality levels and proactively identifies users with insufficient service quality levels due to poor and unstable network connections, which - especially while executing critical tasks such as participating in exams - are to be prioritized in the management of computational and network resources. The modules *Dynamic Computational Resource Manager* and *Network Traffic Scheduler* manage resource allocation, with the aim to accommodate users with disadvantaged network access, by decreasing response times by the platform via increased computational resources and decreasing latency of user requests by increased network resources, such as dedicated bandwidth for prioritized users. The QoS module is present both on the cloud level, as well as on the edge level: QoS management at the edge nodes ensures that computational and network resources are efficiently allocated at the edge level as well, to achieve the required service quality levels. Additionally, edge caching enables storing critical content directly at the edge nodes, physically close to the user’s location to shorten the pathway of network traffic.